IMC 2025: which experiments could have run on JumpServe?

A thought experiment. All 72 papers in the IMC 2025 program, classified by their **primary** data-collection method, read from the abstracts on the program page. One reader, abstracts only, so treat individual placements as judgement calls; many papers mix methods and the note says so where it matters.

	Category	Papers	Share
A	Controlled single-machine experiment: fits the loop as designed, once experiment types are pluggable	11	15%
B	Self-contained active measurement from a cloud vantage point: fits once region is a launch parameter	17	24%
S	Fits the vantage-point model but needs scheduled, repeated runs over weeks to years	6	8%
C	Needs coordinated machines and usually control of routing: multi-machine jobs and beyond	5	7%
E	Analysis of existing or public datasets: runs as a pinned job, the loop adds reproducibility but no data	7	10%
F	Does not fit: proprietary or operator data, physical devices, radios, vehicles or people	26	36%

A. Controlled single-machine experiment: fits the loop as designed, once experiment types are pluggable (11)

- **RemapRoute: Local Remapping of Internet Path Changes**: trace-driven simulation part; the live deployment does not
- **Demystifying the mobile control plane characteristics for ubiquitous connectivity**: control-plane functions under load in a testbed
- **Lazy Eye Inspection: Capturing the State of Happy Eyeballs Implementations**: client testbed; Safari would need Mac hardware
- **The Developer, the RFC, and the Middlebox: An HTTP/2 Compliance Story**: 156 tests against 12 proxies
- **Analyzing Compliance and Complications of Integrating Internationalized X.509 Certificates**: parsing tests on 9 TLS libraries, plus CT-log analysis
- **InternetSim: A Fast and Memory-Efficient Internet-Scale Inter-Domain Routing Simulator**: simulator evaluation; needs a large instance
- **From WebGL to WebGPU: A Reality Check of Browser-Based GPU Acceleration**: GPU kernels; needs GPU instance types
- **Approaching 100% Confidence in Stream Summary through ReliableSketch**: sketch evaluation on traces
- **PipHeap: Approximate Heap in the Pipeline Empowering Network Measurement**: software evaluation only; the hardware pipeline does not fit
- **Assessing LEO Satellite Networks for National Emergency Failover**: model and simulator
- **Decoding DNSSEC Errors at Scale: An Automated DNSSEC Error Resolution Framework using Insights from DNSViz Logs**: repair framework on a purpose-built testbed

B. Self-contained active measurement from a cloud vantage point: fits once region is a launch parameter (17)

- **Canvassing the Fingerprinters: Characterizing Canvas Fingerprinting Use Across the Web**: web crawl
- **CookieGuard: Characterizing and Isolating the First-Party Cookie Jar**: crawl of 20,000 sites
- **Dive into the cloud: Unveiling the (Ab)usage of Serverless Cloud Function in the Wild**: active requests to function URLs; the passive-DNS seed list is external

- **An In-Depth Investigation of Data Collection in LLM App Ecosystems**: crawl of an app ecosystem
- **Somesite I Used To Crawl: Awareness, Agency and Efficacy in Protecting Content Creators From AI Crawlers**: crawl and blocking tests fit; the user study does not
- **How I learned to stop worrying and love IPv6: Measuring the Internet Readiness for DNS over IPv6**: DNS resolution of 10M domains under MTU scenarios
- **The Decentralization Dilemma: Performance Trade-Offs in IPFS and Breakpoints**: client-side IPFS performance
- **Chaos in the Chain: Evaluate Deployment and Construction Compliance of Web PKI Certificate Chain**: TLS scan of the top 1M plus client tests
- **Exploration of the Dynamics of Buy and Sale of Social Media Accounts**: marketplace crawl
- **Decoy Databases: Analyzing Attacks on Public Facing Databases**: 278 honeypots for 20 days: parameterised disposable instances with fixed-schema logs
- **Not All Visitors are Bilingual: A Measurement Study of the Multilingual Web from an Accessibility Perspective**: crawl of 120,000 sites
- **A Permissions Odyssey: A Systematic Study of Browser Permissions on Modern Websites**: crawl for permission use
- **Toward Better Efficiency vs. Fidelity Tradeoffs in Web Archives**: crawler comparison
- **Mirror Mirror on the Wall, which APK Mirror Site is the Largest of Them All?**: metadata crawl of mirror sites
- **Replication: Characterizing MPLS Tunnels over Internet Paths**: traceroute campaign
- **Autonomous Systems under AReST: Advanced Revelation of Segment Routing Tunnels**: traceroute and fingerprinting
- **Measuring the deployment of DNSSEC Bootstrapping Using Authenticated Signals**: DNS scans of zones

S. Fits the vantage-point model but needs scheduled, repeated runs over weeks to years (6)

- **Tracking Internet Disruptions in Ukraine: Insights from Three Years of Active Full Block Scans**: scans every two hours for three years
- **Quantifying the Threat of Sandwiching MEV on Jito: A Measurement of Solana's Leading Validator Client**: three months of continuous collection
- **Attacks Come to Those Who Wait: Long-Term Observations in an SSH Honeynet**: long-term honeynet
- **Measuring the Time Source Vulnerabilities in the NTP Ecosystem**: long-term scans of NTP servers, plus an operator survey
- **On YouTube Search API Use in Research**: identical API queries over 12 weeks
- **Unraveling the Complexities of MTA-STS Deployment and Management in Securing Email**: 31 months of scans, plus a survey

C. Needs coordinated machines and usually control of routing: multi-machine jobs and beyond (5)

- **LACeS: an Open, Fast, Responsible and Efficient Longitudinal Anycast Census System**: needs an anycast deployment of its own plus dispersed probers
- **Time To Scan: Digging into NTP-based IPv6 Scanning**: needs long-lived NTP Pool servers feeding a scanner
- **A Framework to Evaluate MPIC Security using Real-World BGP Announcements**: about 1,500 BGP hijacks of own prefixes observed from 100+ perspectives in 3 clouds
- **A first look into long-lived BGP zombies**: BGP beacons and RIB monitoring for months
- **R&E Routing Policy: Inference and Implication**: BGP combined with active probing

E. Analysis of existing or public datasets: runs as a pinned job, the loop adds reproducibility but no data (7)

- **Sibling Prefixes: Identifying Similarities in IPv4 and IPv6 Prefixes**: DNS resolution data
- **Learning AS-to-Organization Mappings with Borges**: WHOIS and PeeringDB with LLM extraction
- **Prefix2Org : Mapping BGP Prefixes to Organizations**: WHOIS and RPKI
- **Unmasking the Shadow Economy: A Deep Dive into Drainer-as-a-Service Phishing on Ethereum**: on-chain data
- **Replication: A Two Decade Review of Policy Atoms - Tracing the Evolution of AS Path Sharing Prefixes**: BGP collector archives
- **ru-RPKI-ready: the Road Left to Full ROA Adoption**: RPKI and BGP archives
- **Fishing for Smishing: Understanding SMS Phishing Infrastructure and Strategies by Mining Public User Reports**: public user reports

F. Does not fit: proprietary or operator data, physical devices, radios, vehicles or people (26)

- **From Voice to Ads: Auditing Commercial Smart Speakers for Targeted Advertising based on Voice Characteristics**: physical smart speakers
- **Where in the World Are My Trackers? Mapping Web Tracking Flow Across Diverse Geographic Regions**: volunteers' home computers in many countries
- **FP-Inconsistent: Measurement and Analysis of Fingerprint Inconsistencies in Evasive Bot Traffic**: long-lived honey site plus purchased bot traffic
- **The SAP Cloud Infrastructure Dataset: A Reality Check of Scheduling and Placement of VMs in Cloud Computing**: operator dataset
- **Games Are Not Equal: Classifying Cloud Gaming Contexts for Effective User Experience Measurement**: operator traffic
- **Congestion Patterns in a Large-scale RDMA Datacenter**: production datacenter telemetry
- **Patchwork: A Traffic Capture and Analysis Platform for Network Experiments on a Federated Testbed**: a tool for a federated hardware testbed
- **Have you SYN what I see? Analyzing TCP SYN Payloads in the Wild**: network telescope, two years
- **Hello, GenAI? Dissecting Human to Generative AI Calling**: proprietary voice apps on real clients
- **Do Spammers Dream of Electric Sheep? Characterizing the Prevalence of LLM-Generated Malicious Emails**: mail provider's corpus
- **5G Metamorphosis: A Longitudinal Study of 5G Performance from the Beginning**: commercial speed-test dataset
- **Automated, Cross-Layer Root Cause Analysis of 5G Video-Conferencing Quality Degradation**: private 5G and carrier radio telemetry
- **An In-Depth Look into 5G ON-OFF Loops in the Wild**: phones on three carriers
- **Replication: "Performance of Cellular Networks on the Wheels"**: cross-country drive test
- **The Reality of Chasing Shannon's Limit in Optical Wide-Area Networks**: ISP optical data
- **Towards a Non-Binary View of IPv6 Adoption**: observed user traffic
- **Protocol Compliance in Popular RTC Applications**: calls between real app clients
- **Inside Certificate Chains Beyond Public Issuers: Structure and Usage Analysis from a Campus Network**: campus traffic
- **Scrapers selectively respect robots.txt directives: evidence from a large-scale empirical study**: institutional web logs
- **Roam Without a Home: Unraveling the Airalo Ecosystem**: eSIMs on devices in many countries
- **The Anatomy of Olympic Games: a Mobile Traffic Demand Perspective**: production mobile network
- **Unlocking Crowdsourced Propagation Measurements: Accuracy Guarantees for Mobile Phones**: crowdsourced phone radio measurements
- **Fantastic Joules and Where to Find Them. Modeling and Optimizing Router Energy Demand**: physical power measurements of routers
- **From GEO to LEO: First Look Into Starlink In-Flight Connectivity**: 26 flights
- **Satellite IoT in Practice: A First Measurement Study on Network Availability, Performance, and Costs**: 27 ground

stations

- **Understanding and Characterizing Intermediate Paths of Email Delivery: The Hidden Dependencies**: mail provider logs